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Line The sloth bear, an insect-eating animal native to Nepal, exhibits only one behavior that is truly distinct from that of other bear species: The females carry their cubs (at least part time) until the cubs are (5) about nine months old even though the cubs can walk on their own at six months. Cub-carrying also occurs among some other myrmecophagous (ant-eating) mammals; therefore, one explanation is cub-carrying is necessitated by myrmecophagy since (10) myrmecophagy entails a low metabolic rate and high energy expenditure in walking between food patches. However, although polar bears' locomotion is similarly inefficient, polar bear cubs walk along with their mother. Furthermore, the daily movements of sloth (15) bears and American black bears—which are similar in size to sloth bears and have similarly sized home ranges—reveal similar travel rates and distances, suggesting if black bear cubs are able to keep up with their mother, so too should sloth bear cubs.

(20) An alternative explanation is defense from predation. Black bear cubs use trees for defense, whereas brown bears and polar bears, which regularly inhabit treeless environments, rely on aggression to protect their cubs. Like brown bears and polar (25) bears (and unlike other myrmecophagous mammals, which are noted for their passivity), sloth bears are easily provoked to aggression. Sloth bears also have relatively large canine teeth, which appear to be more functional for fighting than for foraging. Like brown (30) bears and polar bears, sloth bears may have evolved in an environment with few trees. They are especially attracted to food-rich grasslands; although few grasslands persist today on the Indian subcontinent, this type of habitat was once widespread there. (35) Grasslands support high densities of tigers, which fight and sometimes kill sloth bears. Sloth bears also coexist with and have been killed by tree-climbing leopards and are often confronted and chased by rhinoceroses and elephants, which can topple trees. (40) Collectively these factors probably selected against tree-climbing as a defensive strategy for sloth bear cubs. Because sloth bears are smaller than brown and polar bears and are under greater threat from dangerous animals, they may have adopted the extra (45) precaution of carrying their cubs. Although cub-carrying may also be adaptive for myrmecophagous foraging, the behavior of sloth bear cubs, which climb on their mother's back at the first sign of danger, suggests that predation was a key stimulus.

Questions 500–504 refer to the passage.

500. The author mentions rhinoceroses and elephants (see line 39) primarily in order to

A. explain why sloth bears are not successful foragers in grassland habitats

B. identify the predators that have had the most influence on the behavior of sloth bears

C. suggest a possible reason that sloth bear cubs do not use tree-climbing as a defense

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D. provide examples of predators that were once widespread across the Indian subcontinent

E. defend the assertion that sloth bears are under greater threat from dangerous animals than are other bear species

501. Which of the following, if true, would most weaken the author's argument in lines 14–19?

A. Cub-carrying behavior has been observed in many non-myrmecophagous mammals.

B. Many of the largest myrmecophagous mammals do not typically exhibit cub-carrying behavior.

C. Some sloth bears have home ranges that are smaller in size than the average home ranges of black bears.

D. The locomotion of black bears is significantly more efficient than the locomotion of sloth bears.

E. The habitat of black bears consists of terrain that is significantly more varied than that of the habitat of sloth bears.

502. Which of the following is mentioned in the passage as a way in which brown bears and sloth bears are similar?

A. They tend to become aggressive when provoked.

B. They live almost exclusively in treeless environments.

C. They are preyed upon by animals that can climb or topple trees.

D. They are inefficient in their locomotion.

E. They have relatively large canine teeth.

503. The author mentions which of the following as evidence for the view that cub-carrying behavior among sloth bears functions primarily as a defense from predation?

A. The relative passivity of sloth bears in comparison with other species of bears

B. The age at which sloth bear cubs can defend themselves from predators

C. The unsuitability of cub-carrying for myrmecophagous foraging

D. The behavior of sloth bear cubs when they first perceive danger

E. The inefficient locomotion of sloth bears and other myrmecophagous animals

504. The primary purpose of the passage is to

A. trace the development of a particular behavioral characteristic of the sloth bear

B. explore possible explanations for a particular behavioral characteristic of the sloth bear

C. compare the defensive strategies of sloth bear cubs to the defensive strategies of cubs of other bear species

D. describe how certain behavioral characteristics of the sloth bear differ from those of other myrmecophagous mammals

E. provide an alternative to a generally accepted explanation of a particular behavioral characteristic of myrmecophagous mammals

Line Biologists have advanced two theories to explain why schooling of fish occurs in so many fish species. Because schooling is particularly widespread among species of small fish, both theories assume that (5) schooling offers the advantage of some protection from predators.

Proponents of theory A dispute the assumption that a school of thousands of fish is highly visible. Experiments have shown that any fish can be seen, (10) even in very clear water, only within a sphere of 200 meters in diameter. When fish are in a compact group, the spheres of visibility overlap. Thus the chance of a predator finding the school is only slightly greater than the chance of the predator (15) finding a single fish swimming alone. Schooling is advantageous to the individual fish because a

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is advantageous to the individual fish because a predator's chance of finding any particular fish swimming in the school is much smaller than its chance of finding at least one of the same group of fish if the fish were dispersed throughout an area.

However, critics of theory A point out that some fish form schools even in areas where predators are abundant and thus little possibility of escaping detection exists. They argue that the school continues to be of value to its members even after detection. They advocate theory B, the "confusion effect," which can be explained in two different ways.

Sometimes, proponents argue, predators simply cannot decide which fish to attack. This indecision supposedly results from a predator's preference for striking prey that is distinct from the rest of the school in appearance. In many schools the fish are almost identical in appearance, making it difficult for a predator to select one. The second explanation for the "confusion effect" has to do with the sensory confusion caused by a large number of prey moving around the predator. Even if the predator makes the decision to attack a particular fish, the movement of other prey in the school can be distracting. The predator's difficulty can be compared to that of a tennis player trying to hit a tennis ball when two are approaching simultaneously.

Questions 505–508 refer to the passage.

505. According to the passage, theory B states that which of the following is a factor that enables a schooling fish to escape predators?

A. The tendency of fish to form compact groups

B. The movement of other fish within the school

C. The inability of predators to detect schools

D. The ability of fish to hide behind one another in a school

E. The great speed with which a school can disperse

506. According to the passage, both theory A and theory B have been developed to explain how

A. fish hide from predators by forming schools

B. forming schools functions to protect fish from predators

C. schooling among fish differs from other protective behaviors

D. small fish are able to make rapid decisions

E. small fish are able to survive in an environment densely populated by large predators

507. According to one explanation of the "confusion effect," a fish that swims in a school will have greater advantages for survival if it

A. tends to be visible for no more than 200 meters

B. stays near either the front or the rear of a school

C. is part of a small school rather than a large school

D. is very similar in appearance to the other fish in the school

E. is medium-sized

508. The author is primarily concerned with

A. discussing different theories

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A. discussing different theories

B. analyzing different techniques

C. defending two hypotheses

D. refuting established beliefs

E. revealing new evidence

Questions 509–563 — Difficulty: Medium

Line In *Winters v. United States* (1908), the Supreme Court held that the right to use waters flowing through or adjacent to the Fort Berthold Indian Reservation was reserved to American Indians by the treaty (5) establishing the reservation. Although this treaty did not mention water rights, the Court ruled that the federal government, when it created the reservation, intended to deal fairly with American Indians by reserving for them the waters without which their (10) lands would have been useless. Later decisions, citing Winters, established that courts can find federal rights to reserve water for particular purposes if (1) the land in question lies within an enclave under exclusive federal jurisdiction; (2) the land has been (15) formally withdrawn from federal public lands—i.e., withdrawn from the stock of federal lands available for private use under federal land use laws—and set aside or reserved; and (3) the circumstances reveal the government intended to reserve water as well as (20) land when establishing the reservation.

Some American Indian tribes have also established water rights through the courts based on their traditional diversion and use of certain waters prior to the United States' acquisition of sovereignty. For (25) example, the Rio Grande pueblos already existed when the United States acquired sovereignty over New Mexico in 1848. Although they at that time became part of the United States, the pueblo lands never formally constituted a part of federal public (30) lands; in any event, no treaty, statute, or executive order has ever designated or withdrawn the pueblos from public lands as American Indian reservations. This fact, however, has not barred application of the Winters doctrine. What constitutes an American Indian (35) reservation is a question of practice, not of legal definition, and the pueblos have always been treated as reservations by the United States. This pragmatic approach is buttressed by *Arizona v. California* (1963), wherein the Supreme Court indicated that (40) the manner in which any type of federal reservation is created does not affect the application to it of the Winters doctrine. Therefore, the reserved water rights of Pueblo Indians have priority over other citizens' water rights as of 1848, the year in which pueblos (45) must be considered to have become reservations.

Questions 509–514 refer to the passage.

509. According to the passage, which of the following was true of the treaty establishing the Fort Berthold Indian Reservation?

A. It was challenged in the Supreme Court a number of times

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A. It was challenged in the Supreme Court a number of times.

B. It was rescinded by the federal government, an action that gave rise to the Winters case.

C. It cited American Indians' traditional use of the land's resources.

D. It failed to mention water rights to be enjoyed by the reservation's inhabitants.

E. It was modified by the Supreme Court in *Arizona v. California*.

510. The passage suggests that, if the criteria discussed in lines 10–20 of the text were the only criteria for establishing a reservation's water rights, which of the following would be true?

A. The water rights of the inhabitants of the Fort Berthold Indian Reservation would not take precedence over those of other citizens.

B. Reservations established before 1848 would be judged to have no water rights.

C. There would be no legal basis for the water rights of the Rio Grande pueblos.

D. Reservations other than American Indian reservations could not be created with reserved water rights.

E. Treaties establishing reservations would have to mention water rights explicitly in order to reserve water for a particular purpose.

511. Which of the following most accurately summarizes the relationship between *Arizona v. California*, as that decision is described in the passage, and the criteria discussed in lines 10–20?

A. *Arizona v. California* abolishes these criteria and establishes a competing set of criteria for applying the Winters doctrine.

B. *Arizona v. California* establishes that the Winters doctrine applies to a broader range of situations than those defined by these criteria.

C. *Arizona v. California* represents the sole example of an exception to the criteria as they were set forth in the Winters doctrine.

D. *Arizona v. California* does not refer to the Winters doctrine to justify water rights, whereas these criteria do rely on the Winters doctrine.

E. *Arizona v. California* applies the criteria derived from the Winters doctrine only to federal lands other than American Indian reservations.

512. The "pragmatic approach" mentioned in lines 37–38 of the passage is best defined as one that

A. grants recognition to reservations that were never formally established but that have traditionally been treated as such

B. determines the water rights of all citizens in a particular region by examining the actual history of water usage in that region

C. gives federal courts the right to reserve water along with land even when it is clear that the government originally intended to reserve only the land

D. bases the decision to recognize the legal rights of a group on the practical effect such a recognition is likely to have on other citizens

E. dictates that courts ignore precedents set by such cases as *Winters v. United States* in deciding what water rights belong to reserved land

513. It can be inferred from the passage that the Winters doctrine has been used to establish which of the following?

A. A rule that the government may reserve water only by explicit treaty or agreement

B. A legal distinction between federal lands reserved for American Indians and federal lands reserved for other purposes

C. Criteria governing when the federal government may set land aside for a particular purpose

D. The special status of American Indian tribes' rights to reserved land

E. The federal right to reserve water implicitly as well as explicitly under certain conditions

514. The author cites the fact that the Rio Grande pueblos were never formally withdrawn from public lands primarily in order to do which of the following?

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primarily in order to do which of the following?

A. Suggest why it might have been argued that the Winters doctrine ought not to apply to pueblo lands

B. Imply that the United States never really acquired sovereignty over pueblo lands

C. Argue that the pueblo lands ought still to be considered part of federal public lands

D. Support the argument that the water rights of citizens other than American Indians are limited by the Winters doctrine

E. Suggest that federal courts cannot claim jurisdiction over cases disputing the traditional diversion and use of water by Pueblo Indians

Line The Black Death, a severe epidemic that ravaged fourteenth-century Europe, has intrigued scholars ever since Francis Gasquet's 1893 study contending that this epidemic greatly intensified the political (5) and religious upheaval that ended the Middle Ages. Thirty-six years later, historian George Coulton agreed but, paradoxically, attributed a silver lining to the Black Death: Prosperity engendered by diminished competition for food, shelter, and work led survivors (10) of the epidemic into the Renaissance and subsequent rise of modern Europe.

In the 1930s, however, Evgeny Kosminsky and other Marxist historians claimed the epidemic was merely an ancillary factor contributing to a general agrarian (15) crisis stemming primarily from the inevitable decay of European feudalism. In arguing that this decline of feudalism was economically determined both Kosminisky and other Marxist historians asserted that the Black Death was a relatively insignificant factor. This became (20) the prevailing view until after World War II, when studies of specific regions and towns revealed astonishing mortality rates ascribed to the epidemic, thus restoring the central role of the Black Death in history.

This central role of the Black Death (traditionally (25) attributed to bubonic plague brought from Asia) has recently been challenged from another direction. Building on bacteriologist John Shrewsbury's speculations about mislabeled epidemics, zoologist Graham Twigg employs urban case studies (30) suggesting the rat population in Europe was both too sparse and insufficiently migratory to have spread plague. Moreover, Twigg disputes the traditional trade-ship explanation for plague transmissions by extrapolating from data on the number of dead rats (35) aboard Nile sailing vessels in 1912. The Black Death, which he conjectures was anthrax instead of bubonic plague, therefore caused far less havoc and fewer deaths than historians typically claim.

Although correctly citing the exacting conditions (40) needed to start or spread bubonic plague, Twigg ignores virtually a century of scholarship contradictory to his findings and employs faulty logic in his single-minded approach to the Black Death. His speculative generalizations about the numbers of (45) rats in medieval Europe are based on isolated studies unrepresentative of medieval conditions, while his unconvincing trade-ship argument overlooks land-based caravans, the overland migration of infected

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rodents, and the many other animals that carry
(50) plague.

Questions 515–516 refer to the passage.

515. The passage is primarily concerned with

A. demonstrating the relationship between bubonic plague and the Black Death

B. interpreting historical and scientific works on the origins of the Black Death

C. employing the Black Death as a case study of disease transmission in medieval Europe

D. presenting aspects of past and current debate on the historical importance of the Black Death

E. analyzing the differences between capitalist and Marxist interpretations of the historical significance of the Black Death

516. Which of the following statements is most compatible with Kosminsky's approach to history as it is presented in the passage?

A. The Middle Ages were ended primarily by the religious and political upheaval in fourteenth-century Europe.

B. The economic consequences of the Black Death included increased competition for food, shelter, and work.

C. European history cannot be studied in isolation from that of the rest of the world.

D. The number of deaths in fourteenth-century Europe has been greatly exaggerated by other historians.

E. The significance of the Black Death is best explained within the context of evolving economic systems.

Line In corporate purchasing, competitive scrutiny is typically limited to suppliers of items that are directly related to end products. With "indirect" purchases (such as computers, advertising, and legal services),
(5) which are not directly related to production, corporations often favor "supplier partnerships" (arrangements in which the purchaser forgoes the right to pursue alternative suppliers), which can inappropriately shelter suppliers from rigorous
(10) competitive scrutiny that might afford the purchaser economic leverage. There are two independent variables—availability of alternatives and ease of changing suppliers—that companies should use to evaluate the feasibility of subjecting suppliers of
(15) indirect purchases to competitive scrutiny. This can create four possible situations.

In Type 1 situations, there are many alternatives and change is relatively easy. Open pursuit of alternatives—by frequent competitive bidding, if
(20) possible—will likely yield the best results. In Type 2 situations, where there are many alternatives but change is difficult—as for providers of employee health care benefits—it is important to continuously test the market and use the results to secure
(25) concessions from existing suppliers. Alternatives provide a credible threat to suppliers, even if the ability to switch is constrained. In Type 3 situations, there are few alternatives, but the ability to switch without difficulty creates a threat that companies
(30) can use to negotiate concessions from existing suppliers. In Type 4 situations, where there are few alternatives and change is difficult, partnerships may be unavoidable.

Questions 517–521 refer to the passage.

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Questions 517–521 refer to the passage.

517. Which of the following can be inferred about supplier partnerships, as they are described in the passage?

A. They cannot be sustained unless the goods or services provided are available from a large number of suppliers.

B. They can result in purchasers paying more for goods and services than they would in a competitive-bidding situation.

C. They typically are instituted at the urging of the supplier rather than the purchaser.

D. They are not feasible when the goods or services provided are directly related to the purchasers' end products.

E. They are least appropriate when the purchasers' ability to change suppliers is limited.

518. Which of the following best describes the relation of the second paragraph to the first?

A. The second paragraph offers proof of an assertion made in the first paragraph.

B. The second paragraph provides an explanation for the occurrence of a situation described in the first paragraph.

C. The second paragraph discusses the application of a strategy proposed in the first paragraph.

D. The second paragraph examines the scope of a problem presented in the first paragraph.

E. The second paragraph discusses the contradictions inherent in a relationship described in the first paragraph.

519. It can be inferred that the author of the passage would be most likely to make which of the following recommendations to a company purchasing health care benefits for its employees?

A. Devise strategies for circumventing the obstacles to replacing the current provider of health care benefits.

B. Obtain health care benefits from a provider that also provides other indirect products and services.

C. Obtain bids from other providers of health care benefits in order to be in a position to negotiate a better deal with the current provider.

D. Switch providers of health care benefits whenever a different provider offers a more competitive price.

E. Acknowledge the difficulties involved in replacing the current provider of health care benefits and offer to form a partnership with the provider.

520. Which of the following is one difference between Type 2 situations and Type 4 situations, as they are described in the passage?

A. The number of alternative suppliers available to the purchaser

B. The most effective approach for the purchaser to use in obtaining competitive bids from potential suppliers

C. The degree of difficulty the purchaser encounters when changing suppliers

D. The frequency with which each type of situation occurs in a typical business environment

E. The likelihood that any given purchase will be an indirect purchase

521. According to the passage, which of the following factors distinguishes an indirect purchase from other purchases?

A. The ability of the purchasing company to subject potential suppliers of the purchased item to competitive scrutiny

B. The number of suppliers of the purchased item available to the purchasing company

C. The methods of negotiation that are available to the purchasing company

D. The relationship of the purchased item to the purchasing company's end product

E. The degree of importance of the purchased item in the purchasing company's business operations

Line Carotenoids, a family of natural pigments, form an important part of the colorful signals used by many animals. Animals acquire carotenoids either

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many animals. Animals acquire carotenoids either directly (from the plants and algae that produce them) or indirectly (by eating insects) and store them in a variety of tissues. Studies of several animal species have shown that when choosing mates, females prefer males with brighter carotenoid-based coloration. Owens and Olson hypothesize that the presence of carotenoids, as signaled by coloration, would be meaningful in the context of mate selection if carotenoids were either rare or required for health. The conventional view is that carotenoids are meaningful because they are rare: healthier males can forage for more of the pigments than can their inferior counterparts. Although this may be true, there is growing evidence that carotenoids are meaningful also because they are required: they are used by the immune system and for detoxification processes that are important for maintaining health. It may be that males can use scarce carotenoids either for immune defense and detoxification or for attracting females. Males that are more susceptible to disease and parasites will have to use their carotenoids to boost their immune systems, whereas males that are genetically resistant will use fewer carotenoids for fighting disease and will advertise this by using the pigments for flashy display instead.

Questions 522–526 refer to the passage.

522. According to the "conventional view" referred to in line 13 of the passage, brighter carotenoid-based coloration in certain species suggests that an individual

A. lives in a habitat rich in carotenoid-bearing plants and insects

B. has efficient detoxification processes

C. has a superior immune system

D. possesses superior foraging capacity

E. is currently capable of reproducing

523. The idea that carotenoid-based coloration is significant partly because carotenoids are required for health suggests that a lack of bright coloration in a male is most likely to indicate which of the following?

A. Inefficient detoxification processes

B. Immunity to parasite infestation

C. Low genetic resistance to disease

D. Lack of interest in mating

E. Lack of carotenoid-storing tissues

524. The passage suggests that relatively bright carotenoid-based coloration is a signal of which of the following characteristics in males of certain animal species?

A. Readiness for mating behavior

B. Ability to fight

C. Particular feeding preferences

D. Recovery from parasite infestation

E. Fitness as a mate

525. The passage implies which of the following about the insects from which animals acquire carotenoids?

A. They do not produce carotenoids themselves

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A. They do not produce carotenoids themselves.

B. They use carotenoids primarily for coloration.

C. They maintain constant levels of carotenoids in their tissues.

D. They are unable to use carotenoids to boost their immune system.

E. They are available in greater abundance than are carotenoid-bearing plants.

526. Information in the passage suggests that which of the following is true of carotenoids that a male animal uses for detoxification processes?

A. They were not acquired directly from plants and algae.

B. They cannot be replenished through foraging.

C. They cannot be used simultaneously to brighten coloration.

D. They do not affect the animal's susceptibility to parasites.

E. They increase the chances that the animal will be selected as a mate.